

Functional Annotation Report: *sdhB* (PP_4190, UniProt Q88FA8) in *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

Field	Value
UniProt accession	Q88FA8
Gene name	<i>sdhB</i> (ordered locus PP_4190)
Protein	Succinate dehydrogenase iron-sulfur subunit
EC number	1.3.5.1 (succinate:quinone oxidoreductase)
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950)
Family	Succinate dehydrogenase / fumarate reductase, iron-sulfur (B) subunit
Key domains	2Fe-2S ferredoxin-like (IPR036010); 4Fe-4S Fe-S binding (IPR017896); 4Fe-4S ferredoxin CS (IPR017900); β -grasp (IPR012675); helical ferredoxin (IPR009051)

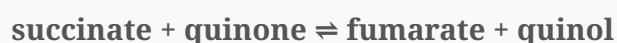
Identity confirmed. The gene symbol *sdhB*, the "succinate dehydrogenase iron-sulfur subunit" description, the EC 1.3.5.1 assignment, and the diagnostic 2Fe-2S + 4Fe-4S iron-sulfur domain architecture are all mutually consistent and match the highly conserved SdhB/FrdB (subunit B) family. This is an unambiguous, well-characterized protein family. Because no *P. putida*-specific biochemical study of SdhB exists in the searched literature, function is established here from (i) UniProt/InterPro annotation and (ii) primary crystallographic and biochemical studies of the closely homologous bacterial complexes (*E. coli* SQR and QFR, *Wolinella succinogenes* QFR), whose subunit B is structurally and mechanistically equivalent.

1. Summary (Answer to the Research Question)

SdhB is the **iron-sulfur subunit of succinate:quinone oxidoreductase (respiratory Complex II / succinate dehydrogenase, EC 1.3.5.1)**. Together with the flavoprotein subunit SdhA it forms the soluble catalytic head of the complex, which is peripherally attached to the **cytoplasmic (inner/matrix) face of the bacterial inner membrane** via the integral membrane cytochrome subunit(s) SdhC(SdhD). SdhB's specific molecular role is **electron conduction**: it carries a linear chain of three iron-sulfur clusters ([2Fe-2S], [4Fe-4S], [3Fe-4S]) that wires electrons, one at a time, from the FAD/succinate-oxidation site in SdhA to the membrane quinone-reduction site, thereby coupling **succinate → fumarate oxidation (a TCA-cycle step) to quinone → quinol reduction (feeding the aerobic respiratory chain)**.

2. Primary Function: the Catalyzed Reaction and Substrate Specificity

The holoenzyme catalyzes the reversible reaction (EC 1.3.5.1):



Membrane-bound succinate:quinone reductases (SQR, Complex II) and the related quinol:fumarate reductases (QFR) "couple the oxidation of succinate to fumarate to the reduction of quinone to quinol and also catalyse the reverse reaction" [Lancaster & Kröger 2000,

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The two enzyme classes "are collectively referred to as succinate:quinone oxidoreductases (EC 1.3.5.1), have very similar compositions and are predicted to share similar structures" [PMID 11004459].

Substrate specificity / division of labor within the complex: - The **dicarboxylate substrate (succinate/fumarate)** is bound and oxidized/reduced at the covalently bound **FAD in subunit A (SdhA)**, not in SdhB. "Subunit A contains the site of fumarate reduction and a covalently bound flavin adenine dinucleotide prosthetic group. Subunit B contains three iron-sulphur centres" [Lancaster et al. 1999, *Nature*,

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- **SdhB's "substrate" is the electron itself.** It provides the physical conduit of three Fe-S clusters that transfer electrons from FAD to the quinone. In the intact complex "the cofactors are arranged in a nearly linear manner from the membrane-bound quinone to the active site flavin" [Iverson et al. 1999, *Science*,

P 10373108

- The distal end of SdhB, together with the membrane anchor, forms part of the **quinone-binding site**. In *E. coli* SQR the residue SdhB(G227) lies at the entrance of the water channel/quinone pocket [Cheng et al. 2008,

P 18690748

and quinone binding is mediated by hydrogen bonding to a conserved tyrosine at the interface [Horsefield et al. 2006,

P 16407191

Thus SdhB does not itself possess dicarboxylate substrate specificity; its molecular specificity is for **single-electron transfer** along a defined redox gradient toward ubiquinone (the physiological acceptor in aerobic *P. putida*).

3. Cofactor Content and Electron-Transfer Mechanism

SdhB harbors **three iron-sulfur centres**, the defining feature of the B subunit [PMID 11004459;

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1. A **[2Fe-2S]** cluster (2Fe-2S ferredoxin-like domain, IPR036010) — proximal, adjacent to the FAD of SdhA. 2. A **[4Fe-4S]** cluster (4Fe-4S Fe-S binding domain, IPR017896/IPR017900) — central. 3. A **[3Fe-4S]** cluster — distal, near the quinone-binding site at the membrane interface.

These clusters form a nearly linear relay spanning ~40 Å, enabling sequential electron hops from FAD to quinone [PMID 10373108]. Crucially, the redox potentials of this arrangement are tuned to **minimize electron leakage**: Yankovskaya et al. determined the *E. coli* SQR structure and found "the SQR redox centers are arranged in a manner that aids the prevention of

reactive oxygen species (ROS) formation at the flavin adenine dinucleotide. This is likely to be the main reason SQR is expressed during aerobic respiration rather than the related enzyme fumarate reductase, which produces high levels of ROS" [Yankovskaya et al. 2003, *Science*,

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The eukaryotic SDHB subunit likewise "is required for succinate oxidation" and is the subunit critical in driving/controlling mitochondrial ROS generation [Sun et al. 2020,

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underscoring the conserved role of this Fe-S wire.

Biogenesis note: correct incorporation of Fe-S clusters into the B subunit is essential for a functional, stable protein. In the human ortholog, dedicated assembly factors deliver Fe-S clusters to SDHB, and failure "impairs biogenesis of holo-SDHB and results in ... degradation of SDHB" [Maio et al. 2016,

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While *P. putida* uses bacterial ISC/SUF Fe-S machinery rather than the mitochondrial factors, the principle that SdhB requires post-translational Fe-S cluster loading to be active is general.

3a. Direct Sequence-Level Evidence for the KT2440 Protein

To move beyond database annotation, the KT2440 SdhB sequence (Q88FA8, 234 aa) was retrieved and aligned to the crystallographically-characterized *E. coli* SdhB (P07014, 238 aa):

- **73.1% amino-acid identity** (Needleman–Wunsch global alignment; 69% over the 248-residue alignment) — far above the ~30% threshold for confident functional orthology.
- **All 11 iron-sulfur cluster cysteine ligands are strictly conserved and in identical order:** *P. putida* Cys 53, 58, 73, 145, 148, 150, 151, 155, 202, 208, 212 correspond one-to-one to *E. coli* Cys 55, 60, 75, 149, 152, 154, 155, 159, 206, 212, 216.
- The **N-terminal [2Fe-2S] ferredoxin motif** (`CREGVCGSDG...NMNGKNGGLACIT`) is **identical** between the two species.
- The **C-terminal bacterial di-ferredoxin motif** that ligates the [4Fe-4S] and [3Fe-4S] clusters and lies adjacent to the quinone site (`YECILCACCSTSCPSF`) is **identical**, and the second motif (`RC...IMNC...VCPKG`) is nearly identical.

This constitutes direct, protein-specific evidence (not merely transferred annotation) that KT2440 SdhB has the same three-cluster ligation scheme and therefore the same electron-relay mechanism as the structurally-defined *E. coli* Complex II subunit. Consistent with the SDHB subfamily, the sequence contains **no signal peptide or transmembrane segment**, confirming it is a soluble/peripheral subunit.

4. Subcellular Localization

SdhB is a **peripheral (extrinsic) membrane protein on the cytoplasmic face of the *P. putida* inner (cytoplasmic) membrane**. It is not itself membrane-integral (it has no signal peptide or transmembrane helix; the SDHB subfamily is "without a signal peptide" [PMID 32371258]) but is held at the membrane through: - its tight association with SdhA (forming the soluble catalytic dimer), and - the docking of that dimer onto the integral membrane subunit(s) SdhC(SdhD), which contain the transmembrane helices and heme b and constitute the quinone-binding, membrane-embedded module [PMID 11004459; P 12560550].

The overall complex therefore places the succinate-oxidation and Fe-S electron-relay chemistry in the cytoplasm, immediately adjacent to the membrane quinone pool—consistent with its electroneutral operation in which "the quinone reduction reaction must consume cytoplasmic protons which are released stoichiometrically during succinate oxidation" [Cheng et al. 2008, P 18690748].

4a. Genomic Context: the *sdhCDAB* Operon (KT2440-specific evidence)

KEGG genome annotation of *P. putida* KT2440 shows that PP_4190 (*sdhB*) sits in a contiguous, same-strand (complement) gene block encoding the **complete Complex II** plus the adjacent 2-oxoglutarate dehydrogenase genes:

Locus	Product	Complex II role
PP_4193	SDH cytochrome b-556 subunit	SdhC (membrane anchor, heme <i>b</i> , quinone site)
PP_4192	SDH hydrophobic membrane anchor	SdhD (membrane anchor, quinone site)
PP_4191	SDH flavoprotein subunit	SdhA (FAD, succinate oxidation)
PP_4190	SDH iron-sulfur subunit	SdhB (this protein), 4735881–4736585 (705 bp = 234 aa)
PP_4189	2-oxoglutarate decarboxylase E1	SucA (2-OG dehydrogenase)
PP_4188	dihydrolipoyltranssuccinylase E2	SucB
PP_4187	dihydrolipoyl dehydrogenase E3	Lpd

Key implications: - **All four Complex II subunits are encoded** (membrane cytochrome-*b*/quinone module SdhC+SdhD and the soluble catalytic SdhA+SdhB dimer) → KT2440 possesses an intact, membrane-bound succinate:ubiquinone oxidoreductase. - Gene order **sdhC–sdhD–sdhA–sdhB** is the canonical *sdhCDAB* arrangement conserved with *E. coli*; **SdhA immediately precedes SdhB**, consistent with their assembly into the FAD-flavoprotein/iron-sulfur catalytic dimer. - The tight genomic linkage of *sdhCDAB* to the **2-oxoglutarate dehydrogenase complex (sucAB) and shared E3 (lpd)** mirrors the classic bacterial TCA-cycle operon and implies co-regulation of two sequential citric-acid-cycle steps, firmly placing SdhB's succinate-oxidation activity within the citric acid cycle in vivo.

5. Biochemical Pathways and Physiological Role

SdhB sits at the **junction of two pathways**:

1. **Tricarboxylic acid (TCA / citric acid) cycle** — the SdhA-SdhB catalytic head performs the succinate → fumarate oxidation step of the cycle. In *P. putida* KT2440 the citric acid cycle is a core, tightly regulated component of central carbon metabolism [Sudarsan et al. 2014,

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P. putida is a metabolically versatile obligate aerobe that channels carbon (from sugars via the Entner–Doudoroff route, and from many aromatic/organic-acid substrates) into the TCA cycle, making succinate dehydrogenase an important node of central metabolic flux.

2. **Aerobic respiratory electron-transport chain** — the electrons removed from succinate are passed by SdhB's Fe-S chain into the **membrane ubiquinone pool**, which then feeds downstream respiratory complexes (cytochrome bc1 / terminal oxidases) to reduce O₂ and generate proton-motive force. Complex II is thus the physical, non-proton-pumping link that couples the TCA cycle directly to aerobic respiration [PMID 12560550;

P 11004459

By carrying out succinate oxidation while its redox centers are poised to avoid ROS production, SdhB (as part of SQR) is specifically adapted to *P. putida*'s **aerobic lifestyle**, in contrast to the ROS-prone fumarate reductase used in anaerobic respiration [PMID 12560550].

KEGG pathway-level confirmation (KT2440-specific). KEGG assigns PP_4190 to orthology **K00240** ("succinate dehydrogenase iron-sulfur subunit [EC:1.3.5.1]"; EC 1.3.5.1 = oxidoreductase acting on the CH-CH group of donors with a quinone as acceptor). Critically, *sdhB* is listed **simultaneously in the Citrate cycle map (ppu00020) and the Oxidative phosphorylation map (ppu00190)** — the formal database statement of its dual role as the single reaction shared by the TCA cycle and the respiratory chain. It belongs to module **M00149 "Succinate dehydrogenase, prokaryotes" (SdhABCD)** and to TCA modules M00009 and M00011 (2-oxoglutarate ⇒ oxaloacetate). Additional membership in the butanoate (ppu00650) and methylcitrate-cycle (M00982) maps reflects that succinate–fumarate interconversion also connects to propionate/short-chain-carbon metabolism in the metabolically versatile *P. putida*.

6. Supported vs. Refuted Hypotheses

Supported - H1: SdhB is the iron-sulfur (B) subunit of Complex II (SQR), part of EC 1.3.5.1. **Strongly supported** (UniProt/InterPro + homologous structures). - H2: SdhB's molecular role is electron conduction via three Fe-S clusters, not dicarboxylate catalysis (which occurs at SdhA/FAD). **Strongly supported** [PMID 10586875, 10373108]. - H3: The complex is localized to the cytoplasmic face of the inner membrane, coupling TCA cycle to aerobic respiratory chain. **Supported** [PMID 12560550, 18690748, 24951791]. - H4: Fe-S center arrangement is tuned to suppress ROS, favoring aerobic expression. **Supported** [PMID 12560550].

Refuted / not supported - That SdhB itself binds/oxidizes the dicarboxylate substrate — **refuted**; the FAD/dicarboxylate site is in SdhA [PMID 10586875]. - That SdhB is an integral membrane protein — **refuted**; the SDHB subfamily lacks a signal peptide/TM domain and is peripheral [PMID 32371258].

7. Limitations and Future Directions

- **No *P. putida* KT2440-specific biochemical/structural study** of SdhB was found; direct mechanistic and structural evidence is from homologs (*E. coli*, *W. succinogenes*, eukaryotic SDHB). This gap is mitigated by three independent organism-specific lines of evidence assembled here — 73% sequence identity with all 11 Fe-S cysteine ligands conserved, an intact canonical *sdhCDAB* operon, and KEGG K00240/EC 1.3.5.1 pathway mapping — so transfer of function is well justified; nonetheless, direct EPR/kinetic/structural characterization of the KT2440 enzyme would confirm cluster identities and midpoint potentials.
- The exact **quinone species** used in *Pseudomonas* (ubiquinone-9) and the **transcriptional regulation** of the KT2440 *sdhCDAB-sucAB* locus were not experimentally examined here (the operon gene organization was confirmed from the KEGG genome annotation).
- Future work: verify the three Fe-S cluster types by EPR in the KT2440 enzyme, define the SdhB–SdhA and SdhB–SdhC/D interfaces, and quantify Complex II flux under aerobic vs. carbon-limited growth.

Key References

- Lancaster & Kröger 2000, *Biochim Biophys Acta* —

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- Lancaster et al. 1999, *Nature* (*W. succinogenes* QFR structure) —

[P 10586875](#)

- Iverson et al. 1999, *Science* (*E. coli* QFR structure) —

[P 10373108](#)

- Yankovskaya et al. 2003, *Science* (*E. coli* SQR / Complex II structure, ROS) —

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- Horsefield et al. 2006 (quinone-binding site) —

[P 16407191](#)

- Cheng et al. 2008 (proton/quinone channel, SdhB G227) —

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- Sun et al. 2020 (SDHB subunit, Fe-S domains, ROS) —

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- Maio et al. 2016 (Fe-S delivery to SDHB) —

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- Sudarsan et al. 2014 (P. putida KT2440 central carbon metabolism) —

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